

Lipofectamine 3000 protocol

15 Sept 2026

- Seed cells in a **12 well plate**
 - Cell type: Caco-2
- No. of cells: _____ (typical $0.5-2 \times 10^5$)
- Cells should be 70–90 % confluent.
- Work in the dark
- Reduce the medium volume in each well to **0.5 mL** to improve transfection efficiency.
- Prepare two mixes
 - Optimem and Lipofectamine (official range 1.5–3 μL per μg DNA, this sheet uses the low end)
 - 50 μL Optimem per well 675 μL
 - 1.5 $\mu\text{L}/\mu\text{g}$ DNA $\rightarrow$ **0.75 μL** Lipofectamine per well 10.13 μL
 - Optimem, DNA and P3000 Reagent – add P3000 last (change the DNA and Lipofectamine and P3000 follow)
 - 50 μL Optimem per well 675 μL
 - 500 ng DNA per well 6750 ng
 - 2 $\mu\text{L}/\mu\text{g}$ DNA $\rightarrow$ **1 μL** P3000 per well 13.5 μL
- Mix well (no need to vortex)
- Add DNA Mix to the Lipofectamine mix
- Incubate 10–15 min and drop-wise **100 μL** mix per well.
- Check cells in microscope
- Spin down plate
- Change medium at the end of the day (2.5 h minimum)
- Incubate cells for 2–4 days at 37 °C

PLASMID / CONDITION	NG/ML	WELLS
<i>EV</i>	<i>823</i>	<i>4.5</i>
<i>WT</i>	<i>718</i>	<i>4.5</i>
<i>MUT</i>	<i>479</i>	<i>4.5</i>
13.5 wells total		

EV 4.5 wells

225 μL Optimem
3.375 μL Lipofectamine

225 μL Optimem
2.73 μL DNA
4.5 μL P3000

WT 4.5 wells

225 μL Optimem
3.375 μL Lipofectamine

225 μL Optimem
3.13 μL DNA
4.5 μL P3000

MUT 4.5 wells

225 μL Optimem
3.375 μL Lipofectamine

225 μL Optimem
4.7 μL DNA
4.5 μL P3000

$$EV: 500 \text{ ng} \times 4.5 \div 823 \text{ ng}/\mu\text{L} = 2.73 \mu\text{L}$$

$$WT: 500 \text{ ng} \times 4.5 \div 718 \text{ ng}/\mu\text{L} = 3.13 \mu\text{L}$$

$$MUT: 500 \text{ ng} \times 4.5 \div 479 \text{ ng}/\mu\text{L} = 4.7 \mu\text{L}$$



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